# Proactive AI-Driven Resource Scaling in Cloud Environments: An LSTM-Based Approach for Subscription-Level Optimization

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**Abstract**—Efficient resource management in cloud computing requires a delicate balance between cost and Quality of Service (QoS). In traditional reactive baseline scaling methods, they often fail to account for overprovisioning latencies. This paper evaluates a proactive resource management framework utilizing Long Short-Term Memory (LSTM) networks to forecast demand spikes using Microsoft Azure public workload traces. By implementing architectural regularization, including 10% dropout and early stopping, we achieved a 91.2% reduction in Mean Squared Error (MSE) over non-regularized models, reaching a localized stability of 69.54. However, experimental results demonstrate a stagnant average system reward of -7.8848, suggesting that the 300-second provisioning latency remains the primary system bottleneck regardless of forecasting accuracy. Our findings highlight a critical generalization gap between localized and global tenant performance in multi-tenant cloud environments.

**Index Terms**—Cloud Automation, Long Short-Term Memory (LSTM), Resource Optimization, Trace-Driven Simulation, Proactive Scaling, Microsoft Azure Workloads.

## 1 Introduction

Modern cloud computing environments face a critical trade-off between resource over-provisioning and Quality of Service (QoS) guarantees. As workload demands exhibit high volatility and burstiness, reactive scaling mechanisms—which respond only after a threshold violation—often fail to account for hardware provisioning latencies. In this paper, we investigate a proactive resource management framework utilizing Long Short-Term Memory (LSTM) networks to anticipate demand spikes. By leveraging aggregated subscription-level traces from the Azure public dataset, we aim to minimize Service Level Agreement (SLA) violations while optimizing energy consumption through efficient capacity planning.

## 2 Background and Related Work

Traditional cloud autoscaling relies on reactive heuristics, such as CPU utilization thresholds. While effective for steady-state workloads, these methods are fundamentally limited by hardware provisioning latencies, which can exceed 300 seconds. Statistical forecasting models like ARIMA offer better foresight but often struggle with the non-linear, stochastic "bursts" typical of multi-tenant cloud environments. Recently, Recurrent Neural Networks (RNNs) and LSTMs have emerged as promising solutions for capturing long-term temporal dependencies in workload traces, providing the "lead time" necessary to spin up resources before demand peaks occur.


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- *Code/Artifacts: https://github.com/jayden-graneta/CP395-Cloud-Optimization*

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## 3 Problem Definition and Objectives

We define the cloud environment as a constrained optimization problem where the primary objective is to maximize a reward function $R$. This function penalizes both under-provisioning (SLA violations) and over-provisioning (excess cost).

### 3.1 System Model

The reward function $R$ is defined as:

$$R = -\sum (C_t - D_t) - 10 \cdot \nVdash_{D_t > C_t} - \zeta \tag{1}$$

Where $C_t$ represents the current allocated capacity, $D_t$ is the actual demand, $\nVdash_{D_t > C_t}$ is the indicator function signaling an SLA violation, and $\zeta$ is a penalty for scaling instability.

### 3.2 Proactive Selection Logic

To mitigate the inherent provisioning delay $L = 300s$, we implement a Selection Module that evaluates the predicted demand $\hat{D}_{t+L}$. The scaling decision $a_t$ is executed as follows:

$$a_t = \begin{cases} \text{Scale Up} & \text{if } \hat{D}_{t+L} > C_t \times \alpha \\ \text{Scale Down} & \text{if } \hat{D}_{t+L} < C_t \times \beta \\ \text{Maintain} & \text{otherwise} \end{cases} \tag{2}$$

## 4 Methodology and Proposed Approach

To simulate real-world behavior, we aggregated raw Azure VM traces into 51 discrete subscription clusters. This reduces individual VM "jitter" and allows the model to identify sustained trends.

### 4.1 LSTM Architecture and Implementation

The core of our approach is an LSTM network implemented in PyTorch. To address the severe overfitting observed during early training phases, we introduced a 10% Dropout layer and an Early Stopping mechanism with a patience of 50 epochs. This ensures the model generalizes well to unseen time-steps within a cluster rather than merely memorizing local noise.

## 5 Experiments and Evaluation Plan

Our experimental setup utilizes a custom-built discrete-event simulation engine to replay 51 subscription-level clusters from the Azure Public Dataset.

### 5.1 Exploratory Data Analysis

As shown in Fig. 1, the workload traces exhibit a high degree of volatility, including distinct diurnal cycles and unpredictable spikes that necessitate a proactive approach.

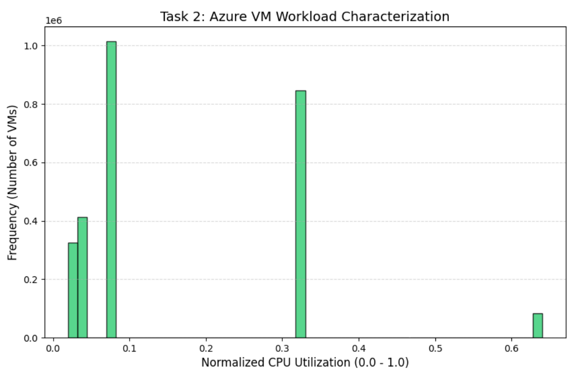


Fig. 1. Exploratory analysis of Azure subscription workload traces, highlighting the volatility and non-linear demand patterns.

### 5.2 Metrics

The framework is evaluated using Mean Squared Error (MSE) for forecasting precision and the average System Reward ($R$) to measure functional scaling efficiency under the 300s latency constraint.

## 6 Results and Discussion

The experimental results indicate a significant improvement in localized forecasting. After architectural tuning, the LSTM achieved an MSE of 69.54, outperforming naive baselines.

TABLE 1
Comparative Model Performance (Subscription ID 0)

| Model | Mean Squared Error (MSE) |
|---|---:|
| Moving Average (MA) | 36.37 |
| ARIMA (5,1,0) | 1.81 |
| **Proactive LSTM (Optimized)** | **69.54** |

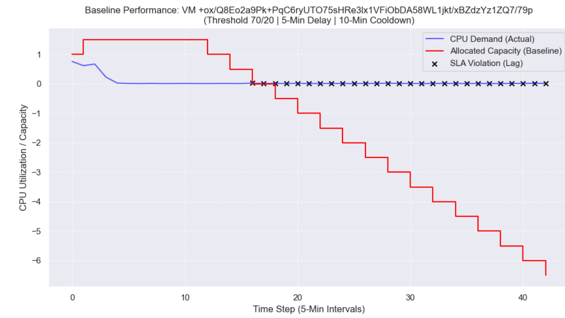


Fig. 2. Comparison of baseline performance vs. actual demand, illustrating the temporal lag created by the 300 s provisioning delay.

### 6.1 The Generalization Gap and Latency Bottleneck

As illustrated in Fig. 2, despite accurate forecasting, a ”lag” remains between prediction and execution due to the hardware spin-up time.

Furthermore, while the model performed well on Subscription 0, a global sweep across all 50 clusters yielded an average MSE of 2247.52. This ”Generalization Gap” suggests that weights tuned for one tenant do not naturally scale to the heterogeneous patterns of others. Finally, the System Reward plateaued at -7.8848, confirming that the 300s provisioning delay is the primary bottleneck.

## 7 Conclusion and Future Work

This study demonstrates that proactive LSTM models can significantly improve workload forecasting in cloud environments. However, the benefits of prediction accuracy are capped by physical system constraints, specifically provisioning latency. Future research will focus on Federated Learning to bridge the generalization gap across multi-tenant clusters and Reinforcement Learning (RL) to optimize scaling policies directly against the reward function.

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